

Multi-Source Candidate Data Transformer — Technical Design (One Pager)

1) Pipeline overview

I use a deterministic 7-step pipeline:

1. **Ingest** input sources (CSV, ATS JSON, notes text). Any source may be missing or malformed.
2. **Extract** source-specific fields into a common intermediate evidence format:
 - `Evidence(field, value, confidence, source, method)`.
3. **Normalize** values:
 - email lowercasing + format check,
 - phone to E.164,
 - country to ISO-3166 alpha-2,
 - date to `YYYY-MM`,
 - skill synonym canonicalization.
4. **Merge / resolve conflicts** into one canonical profile.
5. **Compute confidence** per selected field and `overall\_confidence`.
6. **Project to runtime output schema** using config (`fields`, `from`, `normalize`, include provenance/confidence, `on\_missing`).
7. **Validate** projected output:
 - default schema validated via JSON Schema,
 - custom projection type-checked per config.

This guarantees traceability, robustness, and deterministic outputs for same inputs.

2) Canonical schema + normalization decisions

Canonical record fields follow assignment default:

```
`candidate_id`, `full_name`, `emails[]`, `phones[]`, `location{city,region,country}`,  
`links{linkedin,github,portfolio,other[]}`, `headline`, `years_experience`,  
`skills[{name,confidence,sources[]}]`,  
`experience[{company,title,start,end,summary}]`,  
`education[{institution,degree,field,start_year,end_year}]`,  
`provenance[{field,source,method}]`, `overall_confidence`.
```

Formats chosen:

- Phones: E.164 via `phonenumbers`
- Dates: `YYYY-MM`
- Country: ISO alpha-2 map (fallback pass-through if unknown)
- Skills: canonical lower-case name with synonym mapping (e.g., `ml` -> `machine learning`, `nodejs` -> `node.js`)

3) Merge/conflict policy and confidence

- **Scalar fields** (`full\_name`, `headline`, `years\_experience`, link leaves, location leaves):

choose highest-confidence non-empty evidence.

- **List fields** (`emails`, `phones`, `links.other`):

normalize + dedupe (order preserved).

- **Skills**:

canonical name as key, merge sources, keep max confidence per skill.

- **Experience/Education**:

append normalized records from trusted structures (CSV/ATS); malformed entries ignored.

Confidence scoring:

- Each extractor gives source-aware base confidence (e.g., recruiter email > note regex).
- `overall\_confidence` is average of included field confidences.
- Provenance stores `{field, source, method}` for explainability.

4) Runtime configurable output

Projection layer is separated from canonical merge. Config supports:

- selecting subset of fields,
- remapping (`from`) from canonical paths (supports `phones[0]`, `skills[].name`),
- per-field normalization (`E164`, `canonical`),
- toggling provenance/confidence,
- missing value policy (`null`, `omit`, `error`).

This allows output schema changes with **zero engine code changes**.

5) Edge cases handled + intentional scope limits

Handled:

1. Missing source file -> ignored; pipeline continues.
2. Malformed ATS JSON -> ignored; pipeline continues.
3. Invalid phone/email -> dropped (not invented).
4. Conflicting values across sources -> deterministic winner by confidence.
5. Partial/missing fields -> `null`/`omit`/`error` by config.

Deliberately out of scope (time-boxed):

- full global address parsing/geocoding,
- advanced entity resolution for multiple candidates in one run,
- OCR/PDF resume parsing,
- live API calls/rate-limit logic for GitHub/LinkedIn.